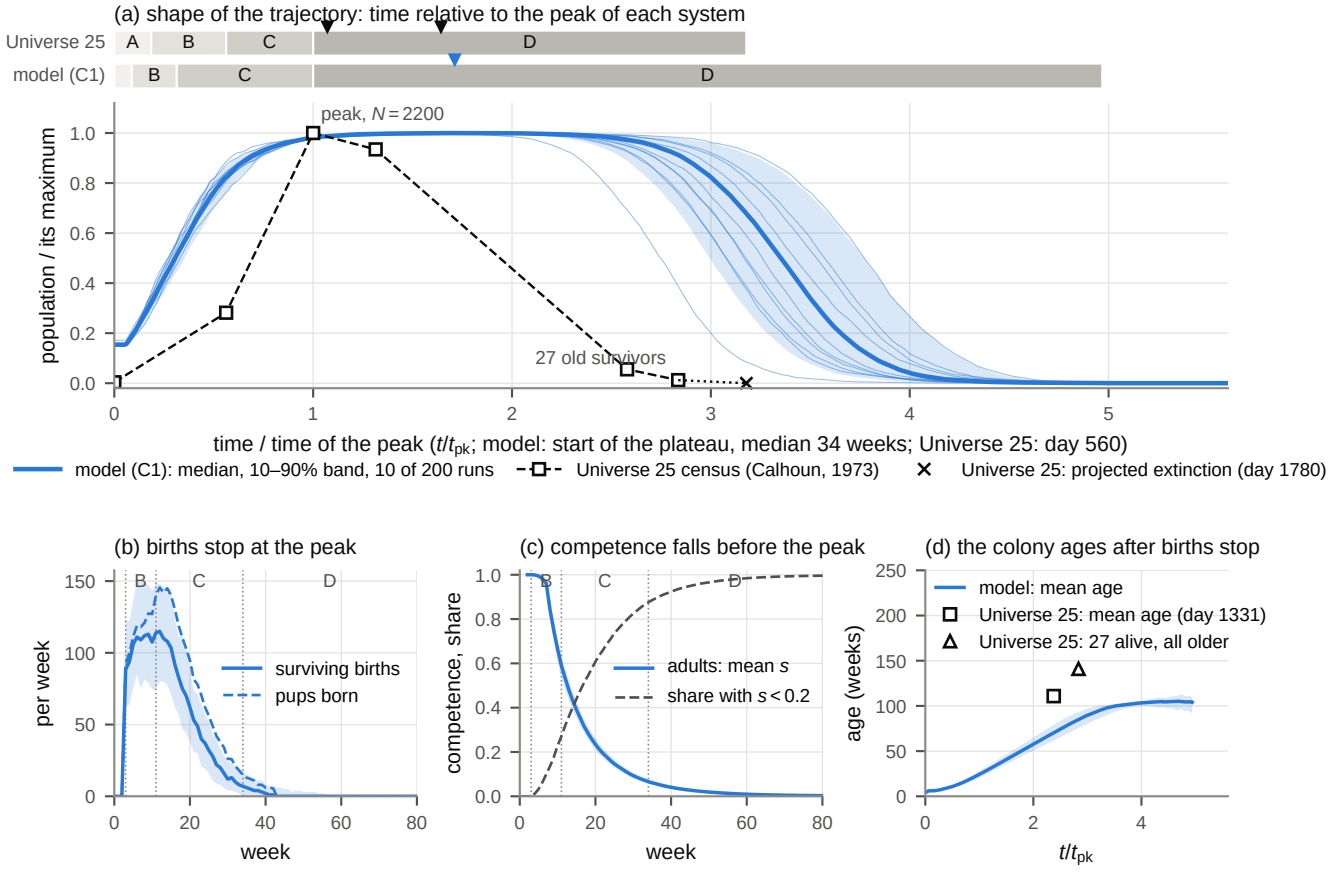

(a) shape of the trajectory: time relative to the peak of each system
Universe 25
A
B
C
D
model (C1)
B
C
D
peak, N = 2200
population / its maximum
1.0
0.8
0.6
0.4
0.2
0.0
27 old survivors
0
1
2
3
4
5
time / time of the peak (t/t_pk; model: start of the plateau, median 34 weeks; Universe 25: day 560)
model (C1): median, 10–90% band, 10 of 200 runs
Universe 25 census (Calhoun, 1973)
Universe 25: projected extinction (day 1780)
(b) births stop at the peak
per week
150
100
50
0
B
C
D
surviving births
pups born
0
20
40
60
80
week
(c) competence falls before the peak
competence, share
1.0
0.8
0.6
0.4
0.2
0.0
B
C
D
adults: mean s
share with s < 0.2
0
20
40
60
80
week
(d) the colony ages after births stop
age (weeks)
250
200
150
100
50
0
model: mean age
Universe 25: mean age (day 1331)
Universe 25: 27 alive, all older
0
2
4
t/t_pk